

Independent Orders and Exact Reductions in Polylogarithm–Zeta Calculus

Entire Hurwitz interval functions, cubic Tornheim directions,
mixed inner orders, and rigidity of Fourier identities

Research continuation prepared for Vladimir Reshetnikov
ProveIt project

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Abstract

We continue the exact-identity program in the ProveIt manuscript and its incoming research reports at revision [5a790187c8e186e41e2b990b4941cb7a1a3c7b6b](#). Four related questions are treated with complete proofs. First, the independent-order harmonic normalization is identified with the established interpolant of Ihara, Nakamura, and Yamamoto. An explicit prefix-residue proof gives its two-endpoint version, joint entire continuation in all orders, composition and shift laws, Gamma generating functions, Stieltjes order jets, and normalized primitives. Second, every admissible third derivative of $T(at, bt, ct)$ at zero is reduced to ordinary zeta derivatives and two specified scalar coordinates: the diagonal derivative Ω and an absolutely convergent logarithmic Gamma series κ . Exact conic and cyclic combinations eliminate the latter; the full boundary germ is also evaluated by a convergent Gamma series. Third, arbitrary weighted colored sums with any mixture of positive and nonpositive integral inner orders admit a terminating finite reduction to nested polylogarithms with one free complex outer order. The construction handles coincident colors, preserves the cumulative alphabet, controls depth, and gives all Laurent constants at nonpositive outer orders by finite local coefficient extraction. Finally, a sampling uniqueness theorem proves the exact finite single-centre reduction criterion for any number of shifted powers, including identities on integer Fourier modes. We distinguish repository advances from known theorems and from unevaluated coordinates. The Gaussian S_6 and revised S_8 candidates remain conjectural. Exact algebra, independent numerical diagnostics, an executable mixed-order compiler, integration notes, and specific further research questions accompany the article.

Keywords. Multiple polylogarithms; Mordell–Tornheim zeta functions; independent spectral orders; truncated multiple zeta interpolation; generalized Stieltjes constants; Gamma functions; harmonic sums; Laurent constants; Fourier uniqueness.

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1 Research scope, status, and conventions

1.1 The source state and the questions answered

The source snapshot is the public ProveIt repository at commit

5a790187c8e186e41e2b990b4941cb7a1a3c7b6b,

inspected on 11 October 2026 (UTC). The two requested directories are `Analysis/Polylogarithms/docs/manuscript` and `docs/incoming` [1, 2]. All eleven incoming archives in that snapshot were inventoried and extracted. The mathematical review concentrated on the relevant harmonic, directional-zeta, mixed-spectral, and Gaussian-certificate sections; this is not a claim to have independently verified every page of the multi-volume manuscript.

The latest incoming reports already contain substantial results. The diagonal cubic digamma finite part is reduced to a third diagonal Tornheim derivative, all quadratic Tornheim directions are known, the all-nonpositive inner-order sector has a finite rational reduction, and common-shift harmonic Laurent formulas and Newton interpolations are available. Repeating those theorems would not address the next questions. Table 1 records the precise continuations here.

Source question	Answer in this article	Status
Independent harmonic exponents (Exact Identities, Question 8)	Entire ordered normalization, explicit relative endpoints and spectral consequences; Section 2	Analytic-normalization component answered; full question remains open
All asymmetric cubic Tornheim rays (directional report, Question 2)	Formula for every admissible direction in two specified scalar coordinates; Section 3	Proved reduction; no independence or ordinary-constant closure claimed
Mixed inner signs at arbitrary depth (directional report, Question 11)	Finite algorithm retaining a free outer order and repeated colors; Section 4	Proved; exact compiler supplied
Scalar identities from integer Fourier samples (mixed-spectral report)	Uniqueness on logarithmically dense samples and complete multicentre criterion; Section 5	Proved in the stated finite power-logarithm class
Short Gaussian S_6 and revised S_8	Search scope and unchanged conjectural status; Section 6	Unresolved

Table 1: Contribution map. “Proved” refers to the stated theorem, not to a claim of literature-wide priority.

The literature reconciliation is substantive. The one-endpoint entire interpolant is precisely the function constructed by Ihara, Nakamura, and Yamamoto [7]; agreement is established by their finite reversed-star formula, not by guessing from integer values. Its identification supplies the independent-order entire normalization requested in one component of Question 8 of the exact-identities report [3]. That question also asks for a smallest generating kernel with a Gamma-separated singular part; neither minimality nor that additional factorization is established here. The additional calculations are presented as exact consequences and useful research organization. The mixed-order and Tornheim theorems are advances relative to the specified source snapshot. The literature search was targeted and does not establish an absolute priority claim for every formula.

1.2 What a reduction means

A reduction is stated relative to a specified output class. A finite sum of multiple polylogarithms with a free complex first index is an exact and useful answer even when it does not collapse to ordinary Gamma and zeta values. Likewise, expressing all cubic directions in Ω and κ is a theorem about a finite spanning collection; it is not a theorem that these coordinates are irreducible or arithmetically independent. In the harmonic sector, the separate order jets retain the nonsymmetric coordinate $\eta(a)$ from the ordered-resonance report [5].

These distinctions prevent three common overstatements: treating a new name for an integral as its evaluation, treating a finite normal form as a minimal basis, and treating equality at special integer parameters as permission to differentiate in a transverse parameter. Every differentiation below is performed on an identity valid in a neighborhood in the variable being differentiated.

1.3 Conventions needed throughout

The superscript $\zeta^{(j)}(s, a)$ denotes differentiation in the first, spectral variable. The generalized Stieltjes constants are fixed by

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j(a)}{j!} u^j, \quad \gamma_0(a) = -\psi(a). \quad (1.1)$$

Write $\gamma_j = \gamma_j(1)$ and $\gamma = \gamma_0$. For a finite upper bound, the strict harmonic convention is

$$H_N(s_1, \dots, s_r) = \sum_{N \geq n_1 > \dots > n_r \geq 1} n_1^{-s_1} \dots n_r^{-s_r}, \quad H_N^{(s)} = H_N(s).$$

The empty word has value one. The local Hurwitz notation in Section 2 starts its shifted summation at zero, and its finite-interval formula specifies the conversion explicitly. Bernoulli numbers use $B_1 = -1/2$ and $te^{xt}/(e^t - 1) = \sum_{n \geq 0} B_n(x)t^n/n!$.

The notation $\text{CT}_{C=c}$ always means the Laurent constant in the displayed affine coordinate $C - c$. It does not mean substitution of individual singular summands. A derivative of a restricted Tornheim ray means that the ray is formed first and its resulting one-variable meromorphic germ is then continued and differentiated. Coherent principal branches are used initially in right half-planes; any extension to other domains is along specified analytic continuation. Normalized primitives include their lower endpoint or their value at the origin, so their additive constants are fixed.

All unrestricted statements are proved analytically or by exact finite algebra in the article. The accompanying computations test conventions, coefficients, and independent evaluation routes. Floating-point agreement is reported as a diagnostic, never as a proof or as an interval-certified error bound.

2 An entire normalization for all independent harmonic exponents

The independent-inner-exponent question in the incoming exact-identities report connects directly to an established interpolant of truncated multiple zeta functions. We identify that interpolant exactly, give a self-contained residue proof of its entire continuation, and develop a relative version with two endpoints. The resulting endpoint, Gamma, and Stieltjes identities give an analytic replacement for the one-parameter Gamma normalization when the inner exponents move independently.

The result does not assert a reduction of every ordered coefficient to ordinary zeta values. In particular the nonsymmetric harmonic Stieltjes coordinate from the incoming ordered-resonance

article remains visible. The analytic continuation and residue structure of multiple Hurwitz zeta functions, and the quasi-shuffle algebra, are classical [8, 9]. The one-endpoint entire interpolation and its stuffle law are due to Ihara, Nakamura, and Yamamoto, with earlier overlapping interpolation work credited in their paper [7]. The relative ordered quotient, residue calculation, and explicit consequences below organize this known construction for the repository's present questions. No literature-wide priority claim is made.

2.1 Definitions and the finite formula

For $\Re a > 0$ and a word $w = (s_1, \dots, s_r)$, let

$$Z_a(w) = \sum_{n_1 > \dots > n_r \geq 0} \prod_{j=1}^r (n_j + a)^{-s_j}, \quad Z_a(\emptyset) = 1. \quad (2.1)$$

Use principal powers. The series is initially normally convergent when $\Re(s_1 + \dots + s_j) > j$ for every j . Its meromorphic continuation is understood below. The deliberately smaller domain $\Re s_j > 1$ for every j suffices for all initial product manipulations.

For $\Re a, \Re b > 0$, define recursively

$$\begin{aligned} \mathcal{H}_{a,b}(\emptyset) &= 1, \\ \mathcal{H}_{a,b}(s_1, \dots, s_r) &= Z_b(s_1, \dots, s_r) - \sum_{j=1}^r Z_a(s_1, \dots, s_j) \mathcal{H}_{a,b}(s_{j+1}, \dots, s_r). \end{aligned} \quad (2.2)$$

The sum includes the final term $Z_a(s_1, \dots, s_r)$. Equivalently,

$$\mathcal{H}_{a,b}(w) = \sum_{q=0}^r (-1)^q \sum_{0=i_0 < i_1 < \dots < i_q \leq r} \left\{ \prod_{\nu=1}^q Z_a(s_{i_{\nu-1}+1}, \dots, s_{i_{\nu}}) \right\} Z_b(s_{i_q+1}, \dots, s_r). \quad (2.3)$$

For $q = 0$ the summand means $Z_b(w)$. There are exactly 2^r terms, counting repetitions before simplification. Thus the definition requires no limiting choice, ray, or finite-part prescription.

The first two cases are

$$h_{a,b}(s) := \mathcal{H}_{a,b}(s) = \zeta(s, b) - \zeta(s, a), \quad (2.4)$$

$$\mathcal{H}_{a,b}(s, t) = Z_b(s, t) - Z_a(s, t) - \zeta(s, a) \{ \zeta(t, b) - \zeta(t, a) \}. \quad (2.5)$$

The second line already shows why subtracting just $Z_a - Z_b$ fails: its residue at $s = 1$ is generally nonzero.

There is also a shorter form in terms of the weakly ordered Hurwitz function $Z_a^\star$, whose inequalities in (2.1) are all weak:

$$\mathcal{H}_{a,b}(w) = \sum_{j=0}^r (-1)^j Z_a^\star(s_j, \dots, s_1) Z_b(s_{j+1}, \dots, s_r). \quad (2.6)$$

The empty prefix has value one. The inverse of the strict ordered series has coefficient $(-1)^j Z_a^\star(s_j, \dots, s_1)$: expand each inverse one-index factor, then reverse the weak order. This proves (2.6) in an absolute-convergence domain and hence meromorphically. Proposition 2.1 of [7] therefore gives the exact identification

$$\mathcal{H}_{a,1}(w) = \Psi(w; a - 1). \quad (2.7)$$

Their Corollary 4.2 already proves that this is entire in all indices; the proof below provides a different route through explicit prefix residues and treats the relative endpoints together.

Theorem 2.1 (Entire relative harmonic interpolation). *For every depth r , the expression (2.3) extends to an entire function of $(s_1, \dots, s_r) \in \mathbb{C}^r$. It is jointly holomorphic with a, b on the right half-plane. Every fixed mixed spectral derivative is consequently an ordinary derivative of one entire function, including where several polar hyperplanes of its separate summands meet.*

If $N \geq 0$ is an integer and $a = b + N$, then

$$\mathcal{H}_{b+N,b}(s_1, \dots, s_r) = \sum_{N > n_1 > \dots > n_r \geq 0} \prod_{j=1}^r (n_j + b)^{-s_j}. \quad (2.8)$$

In particular $\mathcal{H}_{M,1}$ is the strict finite multiple harmonic sum with largest index at most $M - 1$.

2.2 A shift-independent prefix-residue lemma

Write $S_j = s_1 + \dots + s_j$. Introduce the rational-polynomial functions

$$\beta_0(z) = \frac{1}{z-1}, \quad \beta_1(z) = -\frac{1}{2}, \quad \beta_{2q}(z) = \frac{B_{2q}}{(2q)!}(z)_{2q-1} \quad (q \geq 1), \quad \beta_{2q+1}(z) = 0 \quad (q \geq 1). \quad (2.9)$$

Here the B_{2q} are Bernoulli numbers, with $B_1 = -1/2$.

Lemma 2.2 (Prefix residue factorization). *The possible polar divisors of $Z_a(s_1, \dots, s_r)$ are among*

$$s_1 = 1, \quad S_k = k - m \quad (2 \leq k \leq r, \quad m \in \mathbb{Z}_{\geq 0}).$$

At a generic point of each divisor the pole is at most simple. In the normal coordinate $S_k - k + m$, its residue is

$$C_{k,m}(s_1, \dots, s_k) Z_a(s_{k+1}, \dots, s_r), \quad (2.10)$$

where $C_{1,0} = 1$ and, for $k \geq 2$,

$$C_{k,m} = \sum_{\substack{e_1 + \dots + e_{k-1} = m \\ e_i \geq 0}} \prod_{i=1}^{k-1} \beta_{e_i} \left(S_i - (i-1) + \sum_{j < i} e_j \right), \quad \text{restricted to } S_k = k - m. \quad (2.11)$$

The coefficient is independent of a . A vanishing $C_{k,m}$ simply means that the indicated possible divisor is absent.

Proof. Sum first over n_1 , put $x = n_2 + a$, and use a finite Euler–Maclaurin expansion

$$\zeta(z, x+1) = \frac{x^{1-z}}{z-1} - \frac{1}{2}x^{-z} + \sum_{q=1}^Q \frac{B_{2q}}{(2q)!}(z)_{2q-1}x^{-z-2q+1} + R_Q(z, x). \quad (2.12)$$

On any prescribed compact set of z , sufficiently large Q makes the remainder holomorphic there, with a uniform estimate $R_Q(z, n+a) = O((n+1)^{1-\Re z-2Q})$ on compact right-half-plane sets of a . The same is true after fixed parameter derivatives, with logarithmic factors. This follows directly from the integral remainder with a bounded periodic Bernoulli function. In the remaining $r-1$ sums the number and size of the inner terms have at most polynomial growth in n_2 on a compact set of all exponents. Taking Q larger therefore makes the remainder normally summable on that compact.

Each explicit term in (2.12) replaces the first two exponents by $s_1 + s_2 - 1 + e$, with coefficient $\beta_e(s_1)$. Repeat this step in the lower-depth functions. This proves meromorphic continuation by

induction and shows that every possible polar equation is a prefix equation of the displayed form. The pole at $s_1 = 1$ has residue $Z_a(s_2, \dots, s_r)$.

To reach a pole involving exactly the first k indices, merge those indices in $k - 1$ steps. Before step i the current leading exponent is $S_i - (i - 1) + e_1 + \dots + e_{i-1}$. After all merges, the final Hurwitz pole has normal coordinate $S_k - k + e_1 + \dots + e_{k-1}$. The paths producing $S_k = k - m$ are exactly the finite set in (2.11). The remaining indices form the factor $Z_a(s_{k+1}, \dots, s_r)$. All coefficients in this procedure are independent of the common shift.

The only denominators in the merging coefficients arise from proper prefix hyperplanes. Away from intersections, the final pole is simple. Since Q can be chosen arbitrarily large on a prescribed compact, the holomorphic remainders cannot add further local poles. This proves the lemma, including its joint parameter assertion. $\square$

For example,

$$C_{2,0} = \frac{1}{s_1 - 1}, \quad C_{2,1} = -\frac{1}{2}, \quad C_{2,2} = \frac{s_1}{12}, \quad C_{2,3} = 0, \quad C_{3,0} = \frac{1}{(s_1 - 1)(s_1 + s_2 - 2)}.$$

These are residues in a sum-of-indices coordinate; replacing that coordinate by a scaled parameter rescales the residue.

Proof of Theorem 2.1. Induct on the depth. At depth one the common residue of the two Hurwitz functions cancels. Suppose all lower-depth $\mathcal{H}_{a,b}$ are entire. Equation (2.2) shows that the only remaining possible poles are global prefix hyperplanes of the full word. Fix one involving the first k indices and work away from all other hyperplanes. The terms with $j < k$ in (2.2) have no pole there. By Lemma 2.2, the residue of the rest is

$$C_{k,m} \left\{ Z_b(s_{k+1}, \dots, s_r) - \sum_{j=k}^r Z_a(s_{k+1}, \dots, s_j) \mathcal{H}_{a,b}(s_{j+1}, \dots, s_r) \right\} = 0.$$

The bracket is exactly the lower-depth convolution identity (2.2), including the empty-prefix term at $j = k$. Thus every generic polar hypersurface is removable. The locally finite union of their intersections has complex codimension at least two. Hartogs extension removes these remaining intersections; equivalently, one can cancel each linear factor in a local meromorphic denominator. This proves joint entire continuation in the exponents. The same proof with a, b in compact right-half-plane sets gives joint holomorphy.

For the interpolation assertion, take $\Re s_j > 1$. Split the strictly ordered indices for $Z_b(w)$ at N : a uniquely determined initial block lies in $n \geq N$, and the remaining block lies in $n < N$. Hence

$$Z_b(w) = \sum_{j=0}^r Z_{b+N}(s_1, \dots, s_j) \sum_{N > n_{j+1} > \dots > n_r \geq 0} \prod_{\nu=j+1}^r (n_\nu + b)^{-s_\nu}.$$

The triangular recursion identifies the finite sums with $\mathcal{H}_{b+N,b}$. Both sides are entire in all exponents, so the identity extends to $\mathbb{C}^r$. $\square$

2.3 Composition, the stuffle product, and endpoint singularities

For bookkeeping, attach a noncommuting letter $[s]$ to each exponent and form the coefficientwise series

$$F_a = 1 + \sum_{r \geq 1} \sum_{s_1, \dots, s_r} Z_a(s_1, \dots, s_r) [s_1] \cdots [s_r].$$

This notation requires only a finite generating alphabet and its additive closure at any application; it is not a sum over an uncountable set. Concatenation of words gives

$$\mathcal{H}_{a,b} = F_a^{-1} F_b. \quad (2.13)$$

Corollary 2.3 (Exact composition and shifts). *With products interpreted by deconcatenation,*

$$\boxed{\mathcal{H}_{a,b}(w) = \sum_{j=0}^r \mathcal{H}_{a,c}(s_1, \dots, s_j) \mathcal{H}_{c,b}(s_{j+1}, \dots, s_r).} \quad (2.14)$$

Moreover

$$\mathcal{H}_{a+1,b}(w) = \mathcal{H}_{a,b}(w) + a^{-s_1} \mathcal{H}_{a,b}(s_2, \dots, s_r), \quad (2.15)$$

$$\mathcal{H}_{a,b}(w) = \mathcal{H}_{a,b+1}(w) + b^{-s_r} \mathcal{H}_{a,b+1}(s_1, \dots, s_{r-1}). \quad (2.16)$$

All of these identities hold for arbitrary complex exponents.

Proof. The composition formula is $F_a^{-1} F_c F_c^{-1} F_b = F_a^{-1} F_b$. Splitting off the smallest index gives $F_a = F_{a+1} E(a)$, where $E(a) = 1 + \sum_s a^{-s} [s]$. Thus $\mathcal{H}_{a+1,b} = E(a) \mathcal{H}_{a,b}$ and $\mathcal{H}_{a,b} = \mathcal{H}_{a,b+1} E(b)$. Taking coefficients proves the two shift formulas. $\square$

The usual strict stuffle identity also holds without exclusions:

$$h_{a,b}(s) h_{a,b}(t) = \mathcal{H}_{a,b}(s, t) + \mathcal{H}_{a,b}(t, s) + h_{a,b}(s + t). \quad (2.17)$$

More generally $\mathcal{H}_{a,b}$ is a character of the quasi-shuffle algebra whose merged-letter operation is $[s] \circ [t] = [s + t]$. Indeed Z_a and Z_b are characters in the initial domain by splitting equality and inequality cases in a product of sums. This remains true meromorphically. Characters into a commutative algebra are closed under convolution and convolution inversion in the connected quasi-shuffle Hopf algebra [9]; (2.13) therefore gives the assertion. Its coefficients are entire by Theorem 2.1.

Inverting the first shift formula gives the useful finite identity

$$\boxed{\mathcal{H}_{a,b}(s_1, \dots, s_r) = \sum_{j=0}^r (-1)^j a^{-S_j} \mathcal{H}_{a+1,b}(s_{j+1}, \dots, s_r),} \quad S_0 = 0. \quad (2.18)$$

It continues the upper endpoint to the principal slit plane $\mathbb{C} \setminus (-\infty, 0]$, by shifting it finitely many times into the right half-plane. Near $a = 0$ the right side is a finite sum of a^{-S_j} times functions holomorphic in a . Taylor expansion of those functions gives a *convergent* local expansion. Thus the exact branch exponents are prefix sums, and every fixed spectral derivative adds only finitely many powers of $\log a$.

Continuation near $a = -N$, on a specified local branch, follows from

$$\mathcal{H}_{a,b} = E(a)^{-1} E(a+1)^{-1} \cdots E(a+N)^{-1} \mathcal{H}_{a+N+1,b}.$$

At fixed depth this is finite coefficient algebra. Only the factor $E(a+N)^{-1}$ is locally singular, so the branch exponents are sums over consecutive subwords. No essential singularity is introduced. The lower endpoint has the simpler local form (2.16): near $b = 0$ it has one possible singular factor b^{-s_r} .

2.4 Gamma closure of the symmetric integer-index sector

The symmetric-sum theorem for a stuffle character gives

$$\sum_{\sigma \in S_r} \mathcal{H}_{a,b}(s_{\sigma(1)}, \dots, s_{\sigma(r)}) = \sum_{\pi} (-1)^{r-|\pi|} \prod_{B \in \pi} (|B| - 1)! \prod_{B \in \pi} h_{a,b} \left(\sum_{j \in B} s_j \right), \quad (2.19)$$

where π runs over set partitions of $\{1, \dots, r\}$. This is the classical symmetric-sum mechanism applied to the entire character. Repeated labels are counted with their permutation multiplicities; no independence of zeta values is assumed.

In particular, as a convergent germ at $u = 0$,

$$\boxed{\sum_{r \geq 0} \mathcal{H}_{a,b}(\{1\}^r) u^r = \frac{\Gamma(a+u)\Gamma(b)}{\Gamma(a)\Gamma(b+u)}}. \quad (2.20)$$

Its first coefficients, with $h_j = h_{a,b}(j)$, are

$$\mathcal{H}_{a,b}(1,1) = \frac{h_1^2 - h_2}{2}, \quad \mathcal{H}_{a,b}(1,1,1) = \frac{h_1^3 - 3h_1h_2 + 2h_3}{6}, \quad h_1 = \psi(a) - \psi(b).$$

There is a simultaneous finite Gamma formula for every finite alphabet of positive integer exponents. Let $p_1, \dots, p_d \geq 1$, let $M = \max_j p_j$, and let $\rho_1, \dots, \rho_M$ be the roots, with multiplicity, of

$$x^M + \sum_{j=1}^d u_j x^{M-p_j}.$$

Then

$$\boxed{\sum_{r \geq 0} \sum_{j_1, \dots, j_r=1}^d \mathcal{H}_{a,b}(p_{j_1}, \dots, p_{j_r}) u_{j_1} \cdots u_{j_r} = \prod_{\ell=1}^M \frac{\Gamma(a - \rho_\ell)\Gamma(b)}{\Gamma(a)\Gamma(b - \rho_\ell)}}. \quad (2.21)$$

The right side is symmetric in the roots and therefore a holomorphic germ in the u_j , even at colliding roots. It evaluates sums over permutations of the prescribed indices; it does not identify individual ordered words with each other.

To prove (2.21), abelianize the quasi-shuffle series. Its logarithm is the formal difference of $\sum_n \log(1 + \sum_j u_j (n+b)^{-p_j})$ and the corresponding a series. This statement first follows when all independent indices have real part greater than one; the symmetric-sum identity and the entire differences $h_{a,b}$ continue it to the integer alphabet. Each coefficient of the difference converges, including the linear exponent-one coefficient. Factoring the polynomial gives the logarithm

$$- \sum_{m \geq 1} \frac{h_{a,b}(m)}{m} \sum_{\ell=1}^M \rho_\ell^m.$$

The Taylor expansion of $\log \Gamma(a - \rho) - \log \Gamma(a) - \log \Gamma(b - \rho) + \log \Gamma(b)$ is exactly this expression. This proves the identity and also supplies a branch-independent coefficient algorithm.

2.5 Independent logarithmic jets and the ordering coordinate

Put $\gamma_m(x)$ for the generalized Stieltjes constants in

$$\zeta(1+u, x) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m \gamma_m(x)}{m!} u^m.$$

Then

$$h_{a,b}^{(m)}(1) = (-1)^m \{\gamma_m(b) - \gamma_m(a)\}. \quad (2.22)$$

Differentiating (2.17) on the diagonal gives the exact first-jet identity

$$\begin{aligned} (\partial_1 + \partial_2) \mathcal{H}_{a,b}(1, 1) &= \{\psi(a) - \psi(b)\} \{\gamma_1(a) - \gamma_1(b)\} \\ &\quad - \zeta'(2, b) + \zeta'(2, a). \end{aligned} \quad (2.23)$$

The incoming ordered-resonance paper uses the coefficient $\eta(a)$ in

$$Z_a(1 + u, 1) = u^{-2} + \gamma_0(a)u^{-1} + \frac{\gamma_0(a)^2 - \zeta(2, a)}{2} + \eta(a)u + O(u^2).$$

Substitution in (2.5) now gives

$$\partial_1 \mathcal{H}_{a,b}(1, 1) = \eta(b) - \eta(a) + \gamma_1(a) \{\psi(a) - \psi(b)\}, \quad (2.24)$$

$$\partial_2 \mathcal{H}_{a,b}(1, 1) = \eta(a) - \eta(b) - \gamma_1(b) \{\psi(a) - \psi(b)\} - \zeta'(2, b) + \zeta'(2, a). \quad (2.25)$$

These formulas explain the scope of the cancellation theorem precisely. It removes the moving poles and makes the separate logarithmic decorations well defined; it does not remove the nonsymmetric arithmetic coordinate η .

2.6 Finite reductions with a nonpositive inner exponent

One independent-index subfamily does have complete depth-one closure. For $m \geq 0$ and arbitrary complex s ,

$$\boxed{\mathcal{H}_{a,b}(s, -m) = \frac{1}{m+1} \left\{ \sum_{j=0}^{m+1} \binom{m+1}{j} B_{m+1-j} h_{a,b}(s-j) - B_{m+1}(b) h_{a,b}(s) \right\}.} \quad (2.26)$$

All apparent singular values in the separate Hurwitz functions are removable within the h differences. In particular,

$$\begin{aligned} \mathcal{H}_{a,b}(s, 0) &= h_{a,b}(s-1) - b h_{a,b}(s), \\ \mathcal{H}_{a,b}(0, s) &= (a-1) h_{a,b}(s) - h_{a,b}(s-1), \\ \mathcal{H}_{a,b}(s, -1) &= \frac{1}{2} \{h_{a,b}(s-2) - h_{a,b}(s-1) - b(b-1) h_{a,b}(s)\}. \end{aligned} \quad (2.27)$$

For a direct proof, first take $\Re s > m+2$ and use the ordinary finite power-sum identity

$$\sum_{k=0}^{n-1} (k+a)^m = \frac{B_{m+1}(n+a) - B_{m+1}(a)}{m+1}$$

in $Z_a(s, -m)$. Expand $B_{m+1}(n+a)$ in powers of $n+a$. Insert the resulting finite Hurwitz formula, and its b version, into (2.5). Since $h_{a,b}(-m) = \{B_{m+1}(a) - B_{m+1}(b)\}/(m+1)$, the terms involving $B_{m+1}(a)$ cancel. This yields (2.26). Entire continuation gives every s . The second formula in the displayed list also follows from the stuffle identity.

As explicit Gamma–Stieltjes specializations at $b=1$,

$$\begin{aligned} \mathcal{H}_{a,1}(1, 0) &= a-1 - \psi(a) - \gamma, \\ \partial_1 \mathcal{H}_{a,1}(1, 0) &= -\log \Gamma(a) - \gamma_1(a) + \gamma_1, \\ \partial_2 \mathcal{H}_{a,1}(0, 1) &= (a-1) \{\gamma_1(a) - \gamma_1\} + \log \Gamma(a). \end{aligned} \quad (2.28)$$

The derivatives are taken before the indicated index specialization. The identity $h'_{a,1}(0) = -\log \Gamma(a)$ fixes the normalization.

These reductions also have normalized antiderivatives. For a path in the right half-plane from b to a , define

$$P_{a,b}(s) = \int_b^a h_{x,b}(s) dx = (a-b)\zeta(s,b) + \frac{\zeta(s-1,a) - \zeta(s-1,b)}{s-1}. \quad (2.29)$$

The combined right side is entire in s . At $s = 1$ it equals

$$P_{a,b}(1) = \log \Gamma(a) - \log \Gamma(b) - (a-b)\psi(b).$$

Replacing each $h_{a,b}(s-j)$ in (2.26) by $P_{a,b}(s-j)$ gives the uniquely normalized integral $\int_b^a \mathcal{H}_{x,b}(s, -m) dx$. Every spectral derivative commutes with this identity. Thus this independent-index family has a finite calculus in Hurwitz-zeta derivatives, Gamma functions, and generalized Stieltjes constants, with its additive constant fixed.

2.7 Polylogarithmic dictionary

For $|z| < 1$, the interpolation theorem gives

$$\sum_{N \geq 0} \mathcal{H}_{N+1,1}(s_1, \dots, s_r) z^N = \frac{\text{Li}_{s_1, \dots, s_r}(z)}{1-z}, \quad (2.30)$$

where the strict multiple polylogarithm has numerator z^{n_1} . This is an ordinary normally convergent identity, locally in all independent exponents. Each spectral derivative inserts the expected logarithmic factors in the harmonic sums, and the normalized Euler primitive of the numerator raises its first index by one. These classical correspondences now attach to an entire interpolation in the finite harmonic endpoint.

This correspondence also explains the compatibility with the positive-word shuffle calculus in Section 4: finite harmonic endpoint coefficients and nested polylogarithmic words encode the same strict ordering. The independent-order kernel and nonsymmetric-jet problems are collected in Section 7.

3 All cubic Tornheim directions from two scalar coordinates

The incoming *Coincident Stieltjes and Directional Zeta* report [10] proves all quadratic derivatives of $T(at, bt, ct)$ and asks for the entire cubic layer. The following calculation answers that question with two explicit scalar coordinates. One is the already isolated diagonal derivative Ω ; the other is an absolutely convergent logarithmic Gamma series. The result also identifies an exact conic of directions and a cyclic symmetrization in which the second coordinate cancels. These are finite identities; no arithmetic independence of the two coordinates is assumed.

Throughout this section,

$$T(A, B, C) = \sum_{m, n \geq 1} m^{-A} n^{-B} (m+n)^{-C}, \quad L = \log(2\pi),$$

initially in its domain of absolute convergence and subsequently by its meromorphic continuation. Set

$$z_2 = \zeta''(0), \quad z_3 = \zeta'''(0), \quad y_3 = \zeta'''(-1), \quad \Omega = \left. \frac{d^3}{dt^3} T(t, t, t) \right|_{t=0}.$$

The order of restriction is part of the notation. In particular, T is not jointly holomorphic at the origin, whereas the one-variable diagonal is holomorphic there [11]. The Stieltjes convention is

$$\zeta(1+u) = \frac{1}{u} + \sum_{r \geq 0} \frac{(-1)^r \gamma_r}{r!} u^r.$$

3.1 A local remainder with no Gamma factors in its polar part

Lemma 3.1 (Elementary polar model). *There is a unique function $R(A, B, C)$ holomorphic in a neighborhood of $(0, 0, 0)$, symmetric in A, B , for which*

$$T(A, B, C) = \zeta(A)\zeta(B) - \frac{C\zeta(B-1)}{A+C} - \frac{C\zeta(A-1)}{B+C} + CR(A, B, C). \quad (3.1)$$

The identity is meromorphic; each quotient retains its indicated numerator until a restriction has been specified.

Proof. For A, B near zero, Jonquière's expansion [12, §25.12(ii)] gives

$$\text{Li}_A(e^{-x}) = \Gamma(1-A)x^{A-1} + \sum_{j \geq 0} \zeta(A-j) \frac{(-x)^j}{j!}, \quad 0 < x < 2\pi.$$

In the Mellin integral

$$\Gamma(C)T(A, B, C) = \int_0^\infty x^{C-1} \text{Li}_A(e^{-x}) \text{Li}_B(e^{-x}) dx,$$

subtract on $(0, 1)$ the six terms

$$\begin{aligned} S(A, B; x) = & \Gamma(1-A)\Gamma(1-B)x^{A+B-2} + \Gamma(1-A)\zeta(B)x^{A-1} + \Gamma(1-B)\zeta(A)x^{B-1} \\ & - \Gamma(1-A)\zeta(B-1)x^A - \Gamma(1-B)\zeta(A-1)x^B + \zeta(A)\zeta(B). \end{aligned}$$

The difference is normally integrable against x^{C-1} on a sufficiently small common parameter polydisc. Each fixed parameter derivative only adds logarithmic powers. Integrating the six displayed monomials shows that some holomorphic symmetric H satisfies

$$T(A, B, C) = \frac{1}{\Gamma(1+C)} \left\{ \zeta(A)\zeta(B) - \frac{C\Gamma(1-A)\zeta(B-1)}{A+C} - \frac{C\Gamma(1-B)\zeta(A-1)}{B+C} + CH(A, B, C) \right\}. \quad (3.2)$$

The terms with denominators $A+B+C-2$, $A+C-1$, and $B+C-1$ belong to H , since these denominators are nonzero at the origin.

Write $G(C) = 1/\Gamma(1+C)$. Subtract the proposed polar part from (3.2) and divide by C . The resulting expression is

$$\begin{aligned} R(A, B, C) = & G(C)H(A, B, C) + \frac{G(C)-1}{C} \zeta(A)\zeta(B) \\ & - \frac{\Gamma(1-A)G(C)-1}{A+C} \zeta(B-1) - \frac{\Gamma(1-B)G(C)-1}{B+C} \zeta(A-1). \end{aligned}$$

Every quotient is holomorphic. For example, $\Gamma(1-A)G(-A) = 1$, so its numerator is divisible by $A+C$ in the local ring of holomorphic functions. This proves existence and symmetry. Uniqueness follows from (3.1) for $C \neq 0$ and then by analytic continuation. $\square$

The meromorphic continuation and Mellin method are classical [11, 13]; the local form (3.2) also occurs in the incoming report. The change from H to R is useful because all Gamma and Euler constant terms are absorbed into a holomorphic function before the coefficient calculation.

For reference, the complete linear jet is

$$\begin{aligned} R(0, 0, 0) &= \frac{L}{2} + \zeta'(-1), \\ R_A(0) = R_B(0) &= \frac{L^2}{8} - \frac{z_2}{2}, \quad R_C(0) = \frac{\zeta''(-1) - z_2}{2}. \end{aligned} \quad (3.3)$$

These formulas need no additional spectral input. The exact C axis in the next proof determines $R(0)$ and $R_C(0)$. The coefficient of AC in the Euler identity (3.5) gives $2R_A = \zeta'(0)^2 - \zeta''(0)$. As a consequence,

$$\left. \frac{d^2}{dt^2} T(t, t, t) \right|_{t=0} = \frac{L^2}{2} - z_2 - \zeta''(-1) + 4R_A + 2R_C = L^2 - 4z_2.$$

This recovers the incoming quadratic diagonal formula by an independent coefficient calculation, while the new result begins at cubic order.

3.2 The quadratic remainder and the cubic ray theorem

Let all partial derivatives in the following notation be evaluated at $(0, 0, 0)$:

$$p = R_{AA} = R_{BB}, \quad q = R_{AB}, \quad r = R_{AC} = R_{BC}, \quad s = R_{CC}.$$

These four coordinates describe the homogeneous quadratic part of R . Three exact restrictions determine three independent combinations.

Lemma 3.2 (Three exact constraints). *The four second partial derivatives satisfy*

$$\boxed{s = \frac{y_3 - z_3}{3}, \quad p + 2r = -\frac{Lz_2}{2} - z_3, \quad q = \frac{\Omega + 6Lz_2 + 8z_3}{6}.} \quad (3.4)$$

Proof. Counting the $n - 1$ positive pairs with sum n gives

$$T(0, 0, C) = \zeta(C - 1) - \zeta(C).$$

Since $\zeta(0)^2 - 2\zeta(-1) = 5/12$, (3.1) yields

$$CR(0, 0, C) = \zeta(C - 1) - \zeta(C) - \frac{5}{12}.$$

Its cubic coefficient gives $s = (y_3 - z_3)/3$.

On the plane $B = 0$, set

$$E(A, C) = T(A, 0, C) = \sum_{n > m \geq 1} n^{-C} m^{-A}.$$

The ordinary Euler stuffle identity, first in an absolutely convergent region and then meromorphically, is

$$E(A, C) + E(C, A) = \zeta(A)\zeta(C) - \zeta(A + C).$$

Inserting (3.1) gives a holomorphic identity

$$\begin{aligned} CR(A, 0, C) + AR(C, 0, A) &= \zeta(A)\zeta(C) - \zeta(A+C) + \frac{\zeta(A) + \zeta(C)}{2} \\ &\quad + \zeta(-1) + \zeta(A-1) + \zeta(C-1). \end{aligned} \quad (3.5)$$

The coefficient of A^2C on its left is $p/2 + r$. On its right it is $\zeta'(0)z_2/2 - z_3/2 = -Lz_2/4 - z_3/2$. This proves the second constraint.

Finally, on the diagonal (3.1) says

$$T(t, t, t) = \zeta(t)^2 - \zeta(t-1) + tR(t, t, t).$$

Taking third derivatives gives

$$\Omega = -3Lz_2 - z_3 - y_3 + 6p + 6q + 12r + 3s.$$

The first two constraints simplify this to $\Omega = -6Lz_2 - 8z_3 + 6q$, proving the last assertion. $\square$

Theorem 3.3 (Every asymmetric cubic direction). *Let $a, b, c \in \mathbb{C}$ satisfy $(a+c)(b+c) \neq 0$. Form the one-variable meromorphic restriction $W_{a,b,c}(t) = T(at, bt, ct)$ before continuing to $t = 0$. This restriction is holomorphic at zero. Define $\kappa = R_{AA}(0, 0, 0)$. Then*

$$\begin{aligned} \boxed{W_{a,b,c}'''(0)} &= abc\Omega + 3c(a^2 + b^2 - c(a+b))\kappa \\ &\quad - \frac{3}{2}(ab(a+b-4c) + c^2(a+b))Lz_2 \\ &\quad + \left(-\frac{a^3+b^3}{2} + 8abc - 3c^2(a+b) - c^3\right)z_3 \\ &\quad + \left(c^3 - \frac{cb^3}{a+c} - \frac{ca^3}{b+c}\right)y_3. \end{aligned} \quad (3.6)$$

The scalar κ is explicitly given by the absolutely convergent series in (3.10) below.

Proof. On an admissible ray the two polar quotients in (3.1) become constants times holomorphic one-variable zeta functions, so the restriction is holomorphic. Its cubic derivative is

$$\begin{aligned} W_{a,b,c}'''(0) &= -\frac{a^3+b^3}{2}z_3 - \frac{3}{2}ab(a+b)Lz_2 - c\left(\frac{b^3}{a+c} + \frac{a^3}{b+c}\right)y_3 \\ &\quad + 3cp(a^2+b^2) + 6cqab + 6c^2r(a+b) + 3c^3s. \end{aligned}$$

Substitute $p = \kappa$ and (3.4). Collecting the coefficients of $\Omega, \kappa, Lz_2, z_3, y_3$ gives (3.6). $\square$

Thus no unresolved partial derivative of the three-variable Tornheim function remains in the answer. The new information relative to the quadratic theorem is one explicitly specified Gamma-series coordinate in addition to the diagonal coordinate already present in the cubic digamma problem. This is a two-coordinate reduction, not a claim that two algebraically independent constants are necessary.

3.3 A convergent Gamma series and all boundary jets

Put

$$\rho(n) = \log \Gamma(n+1) - \left(n + \frac{1}{2}\right) \log n + n - \frac{L}{2} - \frac{1}{12n}, \quad n \geq 1. \quad (3.7)$$

Stirling's expansion [12, §5.11] gives $\rho(n) = -1/(360n^3) + O(n^{-5})$. Consequently

$$M_j = \sum_{n \geq 1} (\log n)^j \rho(n)$$

converges absolutely for every fixed integer $j \geq 0$. No zeta regularization is used in these sums.

Theorem 3.4 (Boundary germ in Gamma and Stieltjes data). *The Dirichlet series*

$$D(A) = \sum_{n \geq 1} \frac{\rho(n)}{n^A}$$

is holomorphic on $\Re A > -2$. Near $A = 0$ it satisfies

$$R(A, 0, 0) = D(A) - \zeta'(A-1) - \frac{1}{2}\zeta'(A) - \zeta(A-1) + \frac{\zeta(1+A) - 1/A}{12}. \quad (3.8)$$

In particular, for every $j \geq 0$,

$$\partial_A^j R(0, 0, 0) = (-1)^j M_j - \zeta^{(j+1)}(-1) - \frac{1}{2}\zeta^{(j+1)}(0) - \zeta^{(j)}(-1) + \frac{(-1)^j \gamma_j}{12}, \quad (3.9)$$

and the scalar in Theorem 3.3 is

$$\kappa = \sum_{n \geq 1} (\log n)^2 \rho(n) - \zeta'''(-1) - \frac{1}{2}\zeta'''(0) - \zeta''(-1) + \frac{\gamma_2}{12}. \quad (3.10)$$

Proof. The decay of ρ proves normal convergence of D and every fixed order derivative on compact subsets of $\Re A > -2$. For $\Re A > 2$ and C near zero, the representation

$$E(A, C) = \sum_{m \geq 1} m^{-A} \zeta(C, m+1)$$

can be differentiated normally in C . Lerch's formula $\zeta'(0, x) = \log \Gamma(x) - L/2$ [12, (25.11.18)] gives

$$E_C(A, 0) = \sum_{m \geq 1} m^{-A} \left(\log \Gamma(m+1) - \frac{L}{2} \right).$$

Expanding the summand using (3.7), in this convergent half-plane, proves

$$E_C(A, 0) = D(A) - \zeta'(A-1) - \frac{1}{2}\zeta'(A) - \zeta(A-1) + \frac{\zeta(A+1)}{12}.$$

Both sides have meromorphic continuations to a neighborhood of zero. On the other hand, differentiation in C of (3.1) with $B = 0$, for fixed $A \neq 0$, gives

$$E_C(A, 0) = \frac{1}{12A} + R(A, 0, 0).$$

Combining these two identities proves (3.8); its apparent $A = 0$ singularity is removable. Taking j derivatives and using the specified Stieltjes convention proves (3.9) and (3.10). $\square$

The first two members of this hierarchy are completely evaluable:

$$\begin{aligned}\sum_{n \geq 1} \rho(n) &= 2\zeta'(-1) + \frac{L}{4} - \frac{1+\gamma}{12}, \\ \sum_{n \geq 1} (\log n) \rho(n) &= -\frac{L^2}{8} - \zeta''(-1) - \zeta'(-1) - \frac{\gamma_1}{12}.\end{aligned}\tag{3.11}$$

They follow from (3.3) and (3.9). We use these as derived consistency identities and make no separate novelty claim for them.

There is also an exact integral involving an order derivative of the ordinary polylogarithm. In the half-plane $\Re A > -2$,

$$D(A) = \int_0^\infty \left(\frac{1}{e^x - 1} - \frac{1}{x} + \frac{1}{2} - \frac{x}{12} \right) \frac{\text{Li}_A(e^{-x})}{x} dx.\tag{3.12}$$

To see this, the Laplace form of Binet's formula [12, §5.9] gives

$$\rho(n) = \int_0^\infty e^{-nx} \left(\frac{1}{e^x - 1} - \frac{1}{x} + \frac{1}{2} - \frac{x}{12} \right) \frac{dx}{x}.$$

This formula can also be verified by differentiating with respect to n using the standard digamma integral and fixing the constant at $n = +\infty$. Summation and integration commute absolutely on $\Re A > -2$. At the lower endpoint the bracket divided by x is $-x^2/720 + O(x^4)$ and the polylogarithm contributes at worst the integrable singularity required by that half-plane; the upper endpoint is exponentially decaying. Fixed order derivatives add logarithmic factors and remain integrable. Hence

$$M_j = (-1)^j \int_0^\infty \left(\frac{1}{e^x - 1} - \frac{1}{x} + \frac{1}{2} - \frac{x}{12} \right) \partial_A^j \text{Li}_A(e^{-x}) \Big|_{A=0} \frac{dx}{x}.\tag{3.13}$$

Equations (3.10) and (3.13) supply equivalent Gamma-series and polylogarithmic-integral coordinates for the same constant. They also connect the entire boundary derivative hierarchy to the Stieltjes constants. Since $-\partial_s H_n^{(s)} \Big|_{s=0} = \log \Gamma(n+1)$, these identities can equivalently be written with order derivatives of generalized harmonic numbers.

3.4 Exact cancellation families and examples

Corollary 3.5 (A conic of directions). *If an admissible direction obeys*

$$a^2 + b^2 = c(a + b),\tag{3.14}$$

then its third derivative is a finite combination of Ω , $L\zeta''(0)$, $\zeta'''(0)$, and $\zeta'''(-1)$. No Gamma-series coordinate occurs.

For example, the direction $(1, 2, 5/3)$ gives the exact relation

$$W_{1,2,5/3}'''(0) = \frac{10}{3}\Omega - \frac{3}{2}Lz_2 - \frac{403}{54}z_3 - \frac{245}{297}y_3.\tag{3.15}$$

The diagonal and the Euler diagonal are familiar points on the same conic. In particular,

$$W_{1,0,1}'''(0) = -\frac{3}{2}Lz_2 - \frac{9}{2}z_3,$$

as also follows directly from $T(t, 0, t) = (\zeta(t)^2 - \zeta(2t))/2$.

Corollary 3.6 (Cyclic elimination). *Let all pairwise sums $a + b$, $a + c$, $b + c$ be nonzero, and put $e_1 = a + b + c$, $e_2 = ab + bc + ca$, $e_3 = abc$. Then*

$$W'''_{a,b,c}(0) + W'''_{b,c,a}(0) + W'''_{c,a,b}(0) = 3e_3\Omega + 3(9e_3 - e_1e_2)Lz_2 + (-2e_1^3 + 3e_1e_2 + 27e_3)z_3. \quad (3.16)$$

Both κ and $\zeta'''(-1)$ cancel.

Proof. Sum (3.6) cyclically. The κ coefficient is $3\sum_{\text{cyc}} c(a^2 + b^2) - 3\sum_{\text{cyc}} c^2(a + b) = 0$. In the y_3 coefficient, terms sharing the denominator $a + c$ combine to $-b^3$, cancelling the sum of cubes. The remaining two coefficients reduce to the elementary symmetric expressions displayed in (3.16). $\square$

The concrete specialization $(a, b, c) = (1, 2, 3)$ is

$$W'''_{1,2,3}(0) + W'''_{2,3,1}(0) + W'''_{3,1,2}(0) - 18\Omega = -36Lz_2 - 72z_3.$$

Individually, two asymmetric cases read

$$\begin{aligned} W'''_{1,2,3}(0) &= 6\Omega - 36\kappa - \frac{27}{2}Lz_2 - \frac{129}{2}z_3 + \frac{102}{5}y_3, \\ W'''_{2,3,1}(0) &= 6\Omega + 24\kappa - \frac{33}{2}Lz_2 + \frac{29}{2}z_3 - 10y_3. \end{aligned}$$

The Euler direction $(1, 0, 2)$ isolates the additional coordinate:

$$W'''_{1,0,2}(0) = -6\kappa - 6Lz_2 - \frac{41}{2}z_3 + 7y_3. \quad (3.17)$$

Thus one may replace κ by a single Euler double-zeta directional derivative if that basis is preferable. The Gamma series gives a convergent evaluation of that derivative, rather than merely naming it.

For a finite linear combination of admissible rays, the map

$$(a, b, c) \mapsto (abc, 3c(a^2 + b^2 - c(a + b)))$$

lists the coefficients of the two residual coordinates. Any linear combination in the kernel of this explicit map reduces to the displayed ordinary zeta derivatives. This is a sufficient exact elimination criterion. Arithmetic relations between Ω and κ , if found later, could enlarge the actual reducible family.

3.5 Verification and status

The accompanying `check_cubic_rays.py` checks the finite algebra symbolically. It computes the Gamma series using finite Stirling subtraction and evaluates Tornheim functions independently through integrated Jonquière series on $(0, 1)$ and an incomplete-Gamma series on $(1, \infty)$. The reported working precision, finite-difference step, truncations, and discrepancies are recorded in `cubic_rays.json`. The largest discrepancy in the three fully asymmetric cases is less than 2.27×10^{-38} at 65-digit working precision. These numerical diagnostics accompany the proof; the floating-point results are not interval certificates. For orientation,

$$\begin{aligned} \kappa &= 3.6336347857639257352355574698800935822\dots, \\ M_2 &= -0.00064921643050147820901953512402746385\dots \end{aligned}$$

The exact contribution is the cubic directional classification, the explicit boundary germ, and their cancellation consequences. The classical ingredients are Mellin continuation, Euler stuffle, Stirling subtraction, and Lerch's formula. The 2025 preprint of Sathyanarayana and Sharan [13], published in 2026, develops fixed-variable Kronecker limits, Herglotz-type relations, and diagonal special values. Its Section 5 proposes further study of simultaneously varying order limits at points of indeterminacy. The 2026 preprint of Dobrowolski [14] concerns positive-integer symmetrized weights. These are relevant adjacent developments. The simultaneous cubic-origin formula above was not located in the inspected versions, but a comprehensive priority claim would require a broader literature review.

Questions about further reduction of Ω and κ , the quartic jet, and systematic cancellation families are stated in Section 7.

4 A finite reduction for arbitrary mixed integer inner orders

The directional-zeta report reduces two integer inner orders and the arbitrary-depth sector in which every inner order is nonpositive. Its Question 11 asks for a finite nested-polylogarithm reduction at arbitrary depth with both signs present, while the outer order stays complex [6]. We solve that formulation. The construction uses root filtering, shuffle products, rational partial fractions, and finite geometric power sums. Each stage is explicit and accommodates repeated colors. The analytic continuation and the directions in which one may differentiate are part of the statement.

The shuffle algebra and rational manipulation of hyperlogarithms are classical and have powerful general implementations [18]. Here the desired output is a finite expression with one free complex order, positive remaining indices, a controlled depth, and the fixed cumulative root alphabet. The construction and its proof are given in full, so their correctness does not depend on an external symbolic integration package.

4.1 Statement, alphabet, and convergence

For integers a_j , positive integers p_j , and nonzero colors $|X_j| \leq 1$, put

$$Z(C) = \sum_{m_1, \dots, m_d \geq 1} \frac{\prod_{j=1}^d X_j^{m_j} m_j^{-a_j}}{(p_1 m_1 + \dots + p_d m_d)^C}. \quad (4.1)$$

If all colors have modulus less than one, this series is entire in C . For unit colors it initially converges absolutely when, for example,

$$\Re C > d + \sum_j \max(0, -a_j).$$

This sufficient half-plane follows by grouping according to the denominator and using at most $O(n^{d-1})$ tuples of total weight n . Its size is not the research claim.

We use the nested convention

$$\text{Li}_{s_1, \dots, s_r}(z_1, \dots, z_r) = \sum_{n_1 > \dots > n_r \geq 1} \frac{z_1^{n_1} \dots z_r^{n_r}}{n_1^{s_1} \dots n_r^{s_r}}. \quad (4.2)$$

The *cumulative colors* are $\xi_j = z_1 \dots z_j$. The relative colors z_j need not have modulus at most one. If $\max_j |\xi_j| < 1$, the series is locally normally convergent in all its orders: writing $n_j - n_{j+1}$ as positive increments, with $n_{r+1} = 0$, bounds the color product by ρ^{n_1} for some $\rho < 1$. Polynomial growth from the denominators is then harmless.

Theorem 4.1 (Finite mixed-order normal form). *Let $P = \#\{j : a_j > 0\}$ and*

$$\mathcal{A} = \bigcup_{j=1}^d \{\alpha : \alpha^{p_j} = X_j\}.$$

The function (4.1) is a finite linear combination of terms

$$\text{Li}_{C+v, k_2, \dots, k_r}(z_1, \dots, z_r), \quad v \in \mathbb{Z}, \quad k_2, \dots, k_r \in \mathbb{Z}_{>0}, \quad (4.3)$$

whose cumulative colors all belong to $\mathcal{A}$. The coefficients are obtained by finite rational operations in the colors and roots, rational Bernoulli numbers, and integral shuffle multiplicities. One may choose

$$r \leq \begin{cases} d, & P = d, \\ P + 1, & 0 < P < d, \\ 1, & P = 0. \end{cases} \quad (4.4)$$

If all X_j are roots of unity of orders dividing q_j , every cumulative color has order dividing $\text{lcm}_j(p_j q_j)$.

The identity initially holds in the region of absolute convergence and holds meromorphically for every C . If every $X_j \neq 1$, both Z and every final term are entire in C . Every fixed derivative in C is obtained by taking the corresponding derivative with respect to the first index of the multiple polylogarithms in (4.3). The fixed integral orders a_j are not free differentiation variables in this formula.

The depth in (4.4) is a bound for this construction; it is not a minimal-depth or period-independence assertion. In particular, one nonpositive factor can substantially shorten a high-depth input when only a few of the remaining indices are positive.

4.2 Step 1: remove weights and shuffle the positive factors

Let the coefficient transform be

$$\mathcal{M}_C \left[\sum_{n \geq 0} f_n t^n \right] = \sum_{n \geq 1} \frac{f_n}{n^C}.$$

The constant term is omitted. The generator of (4.1) is

$$K(t) = \prod_j \text{Li}_{a_j}(X_j t^{p_j}), \quad Z(C) = \mathcal{M}_C K. \quad (4.5)$$

Root filtering gives, for every integer a ,

$$\text{Li}_a(X t^p) = p^{a-1} \sum_{\alpha^p = X} \text{Li}_a(\alpha t). \quad (4.6)$$

Indeed the sum of n th powers of the roots is zero unless p divides n , and is then $pX^{n/p}$. This also checks the factor p^{a-1} . Expanding the product reduces the proof to unit weights and colors in $\mathcal{A}$.

For a nonzero letter α , use the one-forms $\omega_0 = dt/t$ and $\omega_\alpha = \alpha dt/(1 - \alpha t)$, and define iterated integrals recursively by integrating the leftmost letter against the remaining word. The word $0^{a-1}\alpha$ represents $\text{Li}_a(\alpha t)$ for $a > 0$. Partitioning the product of the integration simplices by the order of their coordinates proves the shuffle product, with multiplicities. Thus the P positive factors give

finitely many words with exactly P nonzero letters. A word with zero-block indices $k_1, \dots, k_P$ and successive nonzero letters $\beta_1, \dots, \beta_P$ represents

$$\text{Li}_{k_1, \dots, k_P}(\beta_1 t, \beta_2 / \beta_1, \dots, \beta_P / \beta_{P-1}). \quad (4.7)$$

All the k_i are positive. This identification follows either by expanding the forms geometrically in the recursive integrals, or by differentiating and fixing the value at $t = 0$. Its cumulative colors are exactly the word letters, all in $\mathcal{A}$.

The nonpositive factors form a rational function $R(t)$, since

$$\text{Li}_{-M}(z) = (z d/dz)^M \frac{z}{1-z} \quad (M \geq 0).$$

Each factor is bounded at infinity, so there is no positive-degree polynomial part in the partial-fraction decomposition

$$R(t) = c_\infty + \sum_{\alpha \in \mathcal{A}} \sum_{h=1}^{M_\alpha} \frac{c_{\alpha,h}}{(1-\alpha t)^h}. \quad (4.8)$$

An upper bound for M_α is the sum of $1 - a_j$ over the nonpositive factors with color α . Even if a numerator cancels some poles, the valid coefficient formula remains

$$c_{\alpha,h} = [u^{M_\alpha-h}] u^{M_\alpha} R((1-u)/\alpha). \quad (4.9)$$

This finite Taylor extraction handles repeated colors directly.

4.3 Step 2: a rational prefactor with one free spectral order

For a nested polylogarithm H as in (4.7), write its relative colors as $\beta_1, \dots, \beta_r$ for this subsection. Clearly

$$\mathcal{M}_C \text{Li}_{k_1, \dots, k_r}(\beta_1 t, \beta_2, \dots, \beta_r) = \text{Li}_{C+k_1, k_2, \dots, k_r}(\beta_1, \dots, \beta_r).$$

Define the rational numbers $b_{h,k}$ by

$$\binom{v+h-1}{h-1} = \prod_{\nu=1}^{h-1} (1+v/\nu) = \sum_{k=0}^{h-1} b_{h,k} v^k. \quad (4.10)$$

Their connection to generalized harmonic numbers is explicit:

$$\prod_{\nu=1}^{h-1} (1+v/\nu) = \exp \left(\sum_{j \geq 1} \frac{(-1)^{j+1} H_{h-1}^{(j)}}{j} v^j \right).$$

Lemma 4.2 (Rational-prefactor transform). *For $h \geq 1$ and $H = \text{Li}_{k_1, \dots, k_r}(\beta_1 t, \beta_2, \dots, \beta_r)$,*

$$\begin{aligned} \mathcal{M}_C[(1-\alpha t)^{-h} H] &= \text{Li}_{C+k_1, k_2, \dots, k_r}(\beta_1, \dots, \beta_r) \\ &+ \sum_{k=0}^{h-1} \sum_{j=0}^k b_{h,k} (-1)^j \binom{k}{j} \text{Li}_{C-k+j, k_1-j, k_2, \dots, k_r}(\alpha, \beta_1/\alpha, \beta_2, \dots, \beta_r). \end{aligned} \quad (4.11)$$

The first inner index $k_1 - j$ is permitted to be nonpositive at this intermediate step.

Proof. Convolution of coefficients gives a sum over $n \geq m \geq 1$ with factor $\alpha^{n-m} \binom{n-m+h-1}{h-1}$. The diagonal $n = m$ is the first term because the binomial coefficient equals one there. For $n > m$, expand (4.10) and $(n-m)^k = \sum_{j=0}^k (-1)^j \binom{k}{j} n^{k-j} m^j$. The new outer color is α and the next is β_1/α , which proves the formula by direct reindexing. $\square$

If $P = 0$, no H is present. The same binomial coefficient gives directly

$$Z(C) = \sum_{\alpha, h} \sum_{k=0}^{h-1} c_{\alpha, h} b_{h, k} \text{Li}_{C-k}(\alpha), \quad (4.12)$$

the depth-one theorem already proved in the source. Thus the new construction agrees exactly with that existing sector.

4.4 Step 3: eliminate every nonpositive inner slot

Lemma 4.3 (Finite elimination, including confluent colors). *Suppose an inner index in (4.2), at position $j \geq 2$, equals $-M$ with $M \geq 0$. Put $U = n_{j-1}$ and $L = n_{j+1}$; for the final position set $L = 0$. If $q = z_j \neq 1$, define*

$$R_h(q) = (q d/dq)^h \frac{1}{1-q}.$$

Then

$$\sum_{n=L+1}^{U-1} q^n n^M = \sum_{h=0}^M \binom{M}{h} R_h(q) \left\{ q^{L+1} (L+1)^{M-h} - q^U U^{M-h} \right\}. \quad (4.13)$$

At $q = 1$ the exact replacement is

$$\sum_{n=L+1}^{U-1} n^M = \frac{B_{M+1}(U) - B_{M+1}(L+1)}{M+1}. \quad (4.14)$$

In either case the resulting multiple polylogarithms have one less index. The cumulative alphabet does not enlarge.

Proof. Apply $(q d/dq)^M$ to the finite geometric sum $(q^{L+1} - q^U)/(1-q)$ and use the product rule. This proves (4.13); the Bernoulli difference identity proves (4.14).

Expand the powers of $L+1$ into powers of L . An upper-end term multiplies the preceding color by q and subtracts a nonnegative integer from its order; a lower-end term does the same to the following color and order. At the last position the lower-end term is a constant. In every case the j th summation disappears. In cumulative colors, the upper-end operation deletes the old $(j-1)$ st cumulative letter, and the lower-end operation deletes the old j th letter. When $q = 1$ those two letters coincide. Hence no new cumulative color is introduced. $\square$

If a neighboring positive index becomes nonpositive, repeat the rule. Depth decreases at every repetition, so the algorithm terminates. The first index can become any C +integer and is never eliminated. This explains both the precise output class and the depth bound.

Completion of Theorem 4.1. Apply (4.6), the finite shuffle expansion, partial fractions, Lemma 4.2, and Lemma 4.3. The positive-only case has depth d ; a rational prefactor initially adds at most one index to depth P ; all subsequent eliminations decrease depth. The rational-only case is (4.12). Alphabet preservation and the root-order statement follow at each stage. The proof so far uses locally normally convergent series for interior cumulative colors, and absolute convergence at unit colors in a sufficiently far right half-plane.

For analytic continuation use

$$Z(C) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \prod_j \text{Li}_{a_j}(X_j e^{-p_j t}) dt. \quad (4.15)$$

It follows by termwise Laplace integration in a common convergent half-plane. If no $X_j = 1$, its kernel is analytic at zero and exponentially decreasing at infinity. Subtracting an arbitrarily long Taylor polynomial near zero gives meromorphic terms $1/(C + n)$, whose poles are canceled by $1/\Gamma(C)$. This proves entire continuation. The same argument applies to every output term: choose an integral positive first index as its base Mellin kernel, and note that its iterated-integral word has no singular letter at $t = 1$. Every cumulative letter lies in $\mathcal{A}$, which contains no 1 when no original $X_j = 1$. Its first-order continuation is therefore entire. If some original $X_j = 1$, the convergent power–logarithm expansions below continue the product kernel meromorphically. Each nested output term has the same type of continuation: a positive-index iterated word has a convergent local power–logarithm expansion at $t = 1$. This follows recursively from its defining integrals, since the only singular endpoint form is $dt/(1 - t)$ and integrating a convergent power series times a finite power of $\log(1 - t)$ preserves that class. Apply the same Mellin subtraction to a positive first-index base word to continue its first order. The identity theorem extends the finite reduction, and differentiation in the free variable C is legitimate. $\square$

Radial continuation in the coefficient variable means $K(t) \mapsto K(\rho t)$; in the original variables this is $X_j \mapsto \rho^{p_j} X_j$, not a common damping of all original colors. At resonant colors the resulting Abel limit need not be an ordinary finite value outside the convergence domain. All formulas there mean the stated meromorphic continuation, with Laurent coefficients taken afterward.

4.5 Local Laurent coefficients at every nonpositive outer order

Let $P_1 = \#\{j : a_j > 0, X_j = 1\}$. The integer limits of Jonquière’s expansion [16, Eq. 25.12.12] give

$$\begin{aligned} \text{Li}_k(e^{-pt}) &= \frac{(-pt)^{k-1}}{(k-1)!} (H_{k-1} - \log p - \log t) \\ &\quad + \sum_{n \geq 0, n \neq k-1} \zeta(k-n) \frac{(-pt)^n}{n!}, \quad k \geq 1, \end{aligned} \quad (4.16)$$

$$\text{Li}_{-M}(e^{-pt}) = \frac{M!}{(pt)^{M+1}} + \sum_{n \geq 0} \zeta(-M-n) \frac{(-pt)^n}{n!}, \quad M \geq 0. \quad (4.17)$$

Here $H_0 = 0$. For $X \neq 1$ the ordinary Taylor expansion is

$$\text{Li}_a(Xe^{-pt}) = \sum_{n \geq 0} \frac{(-p)^n}{n!} \text{Li}_{a-n}(X) t^n. \quad (4.18)$$

Products yield a convergent local power–logarithm expansion

$$\prod_j \text{Li}_{a_j}(X_j e^{-p_j t}) = \sum_{n \geq n_0} \sum_{r=0}^{P_1} \kappa_{n,r} t^n (\log t)^r. \quad (4.19)$$

At any prescribed n , its coefficients require only finitely many terms of (4.16)–(4.18).

Theorem 4.4 (Pole structure and finite local Laurent constants). *Set $g(C) = 1/\Gamma(C)$. Near $C = -n$, the complete polar contribution to $Z(C)$ is*

$$g(C) \sum_{r=0}^{P_1} \frac{(-1)^r r! \kappa_{n,r}}{(C+n)^{r+1}}. \quad (4.20)$$

The difference from (4.20) is $g(C)$ times a holomorphic function. For $n \geq 0$, in particular,

$$\boxed{\text{CT}_{C=-n} Z(C) = \sum_{r=0}^{P_1} \frac{(-1)^r r!}{(r+1)!} \kappa_{n,r} g^{(r+1)}(-n).} \quad (4.21)$$

Possible poles at nonpositive integers have order at most P_1 ; possible poles at positive integers have order at most $P_1 + 1$. There are no other possible poles. Missing local coefficients mean zero.

Proof. Taylor-logarithm subtraction in (4.15) reduces the local terms to

$$\int_0^1 t^{C+n-1} (\log t)^r dt = \frac{(-1)^r r!}{(C+n)^{r+1}}.$$

Arbitrarily high subtraction makes the remainder holomorphic on any prescribed compact set. At $C = -n$ with $n \geq 0$, g has a simple zero. Thus its product with the holomorphic remainder has zero constant coefficient, and the constant of each displayed local term is exactly (4.21). The pole assertions follow from the same zero and from the fact that the powers n are integers. $\square$

Formula (4.21) does not determine all higher regular coefficients. For example, the constant coefficient of $Z'(C)$ generally receives the first coefficient of $g(C)$ times the global holomorphic remainder. This is the same obstruction that prevents differentiating an isolated negative-integer value identity transversely. The finite mixed-order formula, which keeps C free, does determine such derivatives in its stated multiple-polylogarithm class.

Reciprocal-Gamma derivatives in (4.21) are finite polynomials in γ , zeta values, and harmonic numbers. Indeed

$$\frac{1}{\Gamma(-n+u)} = (-1)^n n! u \prod_{j=1}^n (1 - u/j) \exp \left(\gamma u + \sum_{k \geq 2} \frac{(-1)^{k+1} \zeta(k)}{k} u^k \right). \quad (4.22)$$

Only degree $r + 1$ is needed for a local logarithm of degree r .

4.6 Explicit triple identities and a resonant Laurent expansion

Define

$$F(C; z) = \sum_{m,n,k \geq 1} \frac{m z^{m+n+k}}{nk(m+n+k)^C}. \quad (4.23)$$

For $|z| < 1$ it is entire in C ; at $z = 1$ the following original series converges absolutely for $\Re C > 2$. Its finite reduction is

$$\begin{aligned} F(C; z) &= 2 \text{Li}_{C-1,1,1}(z, 1, 1) - 2 \text{Li}_{C-1,1}(z, 1) + 2 \text{Li}_{C,1}(z, 1) \\ &\quad + 2 \text{Li}_{C-1}(z) - 2 \text{Li}_C(z). \end{aligned} \quad (4.24)$$

An independent elementary proof groups by $N = m + n + k$:

$$\sum_{m+n+k=N} \frac{m}{nk} = 2N H_{N-1}(1, 1) - 2(N-1) H_{N-1} + 2(N-1).$$

Here $H_N(1, 1) = \sum_{N \geq j > k \geq 1} (jk)^{-1}$. The formula follows by first putting $v = n + k$, summing $1/(n(v-n)) = 2H_{v-1}/v$, and using $\sum_{v=1}^{N-1} H_{v-1} = (N-1)(H_{N-1} - 1)$. Thus this example tests the general construction independently of the shuffle implementation.

The coefficient generator is $\text{Li}_{-1}(z) \text{Li}_1(z)^2$. Consequently

$$\begin{aligned} F(0; -1) &= -\frac{\log^2 2}{4}, & F(-1; -1) &= -\frac{\log 2}{4}, \\ F(-2; -1) &= \frac{\log^2 2 - \log 2 - 1}{8}, & F(1; -1) &= \frac{\log^2 2}{2} + \log 2 - 1. \end{aligned} \quad (4.25)$$

The first three follow by applying $(z d/dz)^{-C}$ to the rational times logarithm generator at the nonpositive integers. For the last, direct integration gives the stronger functional identity

$$F(1; z) = \int_0^z \frac{\text{Li}_1(u)^2}{(1-u)^2} du = \frac{\text{Li}_1(z)^2 - 2\text{Li}_1(z) + 2}{1-z} - 2. \quad (4.26)$$

More generally $z \partial_z F(C; z) = F(C-1; z)$ and the primitive of $F(C; z)/z$ vanishing at zero is $F(C+1; z)$. Every fixed C -derivative obeys the same identities.

At $z = 1$, the kernel in (4.15) is

$$\text{Li}_{-1}(e^{-t}) \text{Li}_1(e^{-t})^2 = \frac{\log^2 t}{t^2} - \frac{\log t}{t} - \frac{\log^2 t}{12} + \frac{\log t}{12} + \frac{1}{4} + O(t(1 + |\log t|^2)).$$

Using (4.20) and (4.22) therefore proves the exact Laurent expansion

$$F(C; 1) = -\frac{1}{6C^2} - \frac{2\gamma + 1}{12C} + \frac{1}{4} - \frac{\gamma^2 + \gamma}{12} + \frac{\zeta(2)}{12} + O(C). \quad (4.27)$$

This example has a genuine double pole at a nonpositive outer order. The finite part is a short Gamma-zeta polynomial, while the whole function is the depth-three expression (4.24).

A second example exercises two different nonpositive-slot eliminations:

$$\begin{aligned} \sum_{m,n,k \geq 1} \frac{k^2 z^{m+n+k}}{mn(m+n+k)^C} &= 2\text{Li}_{C-2,1,1}(z, 1, 1) - 3\text{Li}_{C-2,1}(z, 1) + 3\text{Li}_{C-1,1}(z, 1) \\ &\quad + \frac{7}{2}\text{Li}_{C-2}(z) - \frac{9}{2}\text{Li}_{C-1}(z) + \text{Li}_C(z). \end{aligned} \quad (4.28)$$

Its generator is $\text{Li}_{-2}(z) \text{Li}_1(z)^2$. At $z = 1$ its finite value at $C = 0$ is $-1/12$. At $C = -1 + u$ its Laurent expansion is

$$\frac{1}{60u^2} + \frac{\gamma/60 - 13/720}{u} + \frac{\gamma^2}{120} - \frac{13\gamma}{720} - \frac{\pi^2}{720} - \frac{1}{480} + O(u). \quad (4.29)$$

For verification, the t^1 coefficient of the local kernel is $-\log^2 t/120 - \log t/720 + 1/288$; insertion into (4.20) at $C = -1$ gives (4.29) directly.

4.7 A convergent evaluation and exact pole classification

Write $U(C; z)$ for the left side of (4.28). At $z = 1$ its defining series converges absolutely for $\Re C > 3$. Grouping by $N = m + n + k$ gives the additional finite harmonic identity

$$\begin{aligned} \sum_{m+n+k=N} \frac{k^2}{mn} &= N^2(H_{N-1}^2 - H_{N-1}^{(2)}) - 3N(N-1)H_{N-1} \\ &\quad + \frac{7N^2 - 9N + 2}{2}. \end{aligned} \quad (4.30)$$

This follows by taking the outer coefficient in (4.28), including the vanishing cases $N = 1, 2$. The following ordinary value uses only classical height-one multiple zeta identities, whose beta-integral proof is included to fix the normalization.

Corollary 4.5 (A convergent zeta evaluation).

$$\boxed{\sum_{m,n,k \geq 1} \frac{k^2}{mn(m+n+k)^4} = \frac{\pi^4}{24} + \frac{7\pi^2}{12} - \frac{15}{2}\zeta(3).} \quad (4.31)$$

Proof. Set $C = 4, z = 1$ in (4.28), and use

$$\zeta(2, 1, 1) = \zeta(4), \quad \zeta(2, 1) = \zeta(3), \quad \zeta(3, 1) = \frac{1}{4}\zeta(4).$$

For a self-contained verification of these three classical relations, consider the classical Aomoto–Drinfel’d height-one generating identity, recorded in Kaneko and Ohno [15, Eq. (5)]:

$$\sum_{r,s \geq 1} \zeta(r+1, \{1\}^{s-1}) x^r y^s = 1 - \frac{\Gamma(1-x)\Gamma(1-y)}{\Gamma(1-x-y)}. \quad (4.32)$$

Indeed the shuffle identity gives $(-\log(1-t))^s/s! = \text{Li}_{1,\dots,1}(t, 1, \dots, 1)$, with s indices. Expand this absolutely convergent power series and use $\int_0^1 t^{n-1}(-\log t)^{r-1} dt = (r-1)!/n^r$. Termwise integration is justified by nonnegative summands and gives

$$\zeta(r+1, \{1\}^{s-1}) = \frac{1}{(r-1)!s!} \int_0^1 \frac{(-\log t)^{r-1}(-\log(1-t))^s}{t} dt.$$

Summing on a sufficiently small bidisc turns the right side into

$$x \int_0^1 t^{-x-1} ((1-t)^{-y} - 1) dt = 1 + x \frac{\Gamma(-x)\Gamma(1-y)}{\Gamma(1-x-y)}.$$

The beta evaluation is first justified for $\Re x < 0$ and then by analytic continuation near zero. The functional equation of Gamma gives (4.32). Finally

$$\log \frac{\Gamma(1-x)\Gamma(1-y)}{\Gamma(1-x-y)} = \sum_{j \geq 2} \frac{\zeta(j)}{j} (x^j + y^j - (x+y)^j).$$

Its xy^2 , xy^3 , and x^2y^2 coefficients give the three displayed identities; in the last case use $\zeta(2)^2 = 5\zeta(4)/2$. Formula (4.28) now gives $15\zeta(4)/4 + 7\zeta(2)/2 - 15\zeta(3)/2$, which is (4.31). $\square$

4.8 The full pole pattern at the resonant color

Write $U(C) = U(C; 1)$ and let B_j denote the Bernoulli numbers with $B_1 = -1/2$. For $\Re C > 3$,

$$U(C) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} K(t) dt, \quad K(t) = \text{Li}_1(e^{-t})^2 \text{Li}_{-2}(e^{-t}). \quad (4.33)$$

The integral follows directly from the Gamma integral for the denominator; absolute convergence follows from the coefficient bound $O(N^2(1 + \log N)^2)$.

Proposition 4.6 (Actual spectral poles). *The meromorphic function U has a triple pole at $C = 3$, double poles at $C = 2, 1$ and at every negative odd integer, and simple poles at every negative even integer. These are all its poles. The point $C = 0$ is regular and*

$$U(0) = -\frac{1}{12}. \quad (4.34)$$

For every integer $m \geq 1$,

$$\operatorname{Res}_{C=-2m} U(C) = \frac{m}{m+1} B_{2m+2}. \quad (4.35)$$

For every integer $m \geq 0$, the coefficient of the leading double pole at $C = -(2m+1)$ is

$$[(C+2m+1)^{-2}] U(C) = -\frac{B_{2m+4}}{m+2}. \quad (4.36)$$

Proof. The convergent local expansions for $0 < t < 2\pi$ are

$$\begin{aligned} \operatorname{Li}_1(e^{-t}) &= -\log t + a(t), & a(t) &= \frac{t}{2} - \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} t^{2r}, \\ \operatorname{Li}_{-2}(e^{-t}) &= b(t), & b(t) &= \frac{2}{t^3} + \sum_{r \geq 1} \frac{B_{2r+2}}{(2r+2)(2r-1)!} t^{2r-1}. \end{aligned}$$

The second expansion follows by differentiating $(e^t - 1)^{-1}$ twice. The first follows by integrating its negative and fixing the constant from $\operatorname{Li}_1(e^{-t}) + \log t \rightarrow 0$. Thus

$$K(t) = b(t)(\log t)^2 - 2a(t)b(t)\log t + a(t)^2 b(t).$$

Mellin subtraction shows that a coefficient $\kappa_{n,r} t^n (\log t)^r$ contributes the singular model

$$\frac{1}{\Gamma(C)} \frac{(-1)^r r! \kappa_{n,r}}{(C+n)^{r+1}}.$$

All other contributions are holomorphic locally, after a sufficiently long Taylor–logarithm subtraction. There are no additional poles because the tail of the integral decays exponentially.

At the first four powers,

$$K(t) = \frac{2 \log^2 t}{t^3} - \frac{2 \log t}{t^2} + \frac{\log t/6 + 1/2}{t} - \frac{1}{12} + O(t \log^2 t). \quad (4.37)$$

The three positive poles therefore have orders three, two, and two, with nonzero leading coefficients 2, 2, and $-1/6$, respectively. The zero of $1/\Gamma(C)$ at $C = 0$ cancels the simple local Mellin pole there; its derivative at zero is one, proving (4.34).

For a positive even n , $b(t)$ has no t^n term. Since the only odd power in $a(t)$ is $t/2$, the coefficient of $t^n \log t$ in K is $-b_{n-1}$, where

$$b_{n-1} = \frac{B_{n+2}}{(n+2)(n-1)!}.$$

Using $(1/\Gamma)'(-n) = (-1)^n n!$ gives residue $n! b_{n-1} = n B_{n+2}/(n+2)$, which is (4.35). For positive odd n , the coefficient of $t^n \log^2 t$ is $b_n = B_{n+3}/((n+3)n!)$. The same reciprocal-Gamma zero gives leading double-pole coefficient $-2n! b_n$, proving (4.36). The even-index Bernoulli numbers appearing here are nonzero, so these are actual poles, of exactly the stated orders. $\square$

At the next nonpositive even order one also obtains

$$\operatorname{CT}_{C=-2} U(C) = \frac{19}{720} - \frac{\gamma}{60}. \quad (4.38)$$

Indeed $[t^2]K(t) = \log(t)/120 + 1/1440$, and $1/\Gamma(-2+u) = 2u + (2\gamma-3)u^2 + O(u^3)$. Multiplication by its local Mellin model gives $2/1440 - (2\gamma-3)/120$, proving the displayed constant. This is a Laurent constant of the continuation, not a value of the positive-term defining series at $C = -2$.

4.9 The depth-one boundary: Gamma, polygamma, and Stieltjes data

When the normal form reaches depth one at a q th root of unity α , finite residue-class summation gives

$$\mathrm{Li}_s(\alpha) = q^{-s} \sum_{j=1}^q \alpha^j \zeta(s, j/q). \quad (4.39)$$

For $s \neq 1$ its k th order derivative is

$$q^{-s} \sum_{j=1}^q \alpha^j \sum_{\ell=0}^k \binom{k}{\ell} (-\log q)^{k-\ell} \zeta^{(\ell)}(s, j/q). \quad (4.40)$$

For nontrivial α the common pole at $s = 1$ cancels first, so the finite formula there is

$$\mathrm{Li}_1^{[k]}(\alpha) = \frac{1}{q} \sum_{j=1}^q \alpha^j \sum_{\ell=0}^k \binom{k}{\ell} (-\log q)^{k-\ell} (-1)^\ell \gamma_\ell(j/q). \quad (4.41)$$

These classical identities, already used in the source, fix the normalization of the depth-one output. Lerch's identity $\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ and $\psi^{(r)}(a) = (-1)^{r+1} r! \zeta(r+1, a)$, $r \geq 1$, give the corresponding Gamma and polygamma expressions [17]. The new theorem does not erase the positive-depth order derivatives left by its normal form; further reductions require additional identities.

5 Sampled identities and finite single-centre reduction

The mixed-spectral report proves a two-centre functional criterion and explicitly asks whether identities on integer Fourier modes imply the corresponding scalar identities [4, Section 6, question on sampled identities]. In the finite power–logarithm class used there, the answer is yes. We give an elementary proof that also works on many sparser sets, then obtain the exact finite-reduction criterion for any number of distinct twists.

Classical uniqueness theorems, such as Carlson's theorem, concern much larger holomorphic function classes [19]. The elementary lemma below uses the convergent expansions specific to the present functions. The contribution to the repository is the complete sampling implication and its application to finite twisted-family reduction; no new general Carlson theorem is claimed.

Definition 5.1 (Finite power–logarithm germs at infinity). Let $\mathcal{P}$ consist of functions, for sufficiently large positive x , of the form

$$f(x) = \sum_{\nu=1}^J x^{\alpha_\nu} \sum_{k=0}^{K_\nu} (\log x)^k h_{\nu k}(1/x), \quad (5.1)$$

where the $h_{\nu k}$ are holomorphic in a disc about zero and $\alpha_\nu \in \mathbb{C}$. Powers of positive x use the real logarithm. Every finite product or sum of shifted powers and nonnegative integral powers of their logarithms belongs to this class.

For the last assertion, expand $(x+a)^s = x^s(1+a/x)^s$ and $\mathrm{Log}(x+a) = \log x + \mathrm{Log}(1+a/x)$ for $x > 2|a|$. The expansions are convergent, rather than merely asymptotic.

Theorem 5.2 (Uniqueness on an asymptotically dense logarithmic mesh). *Let $f \in \mathcal{P}$. Suppose that $x_j \rightarrow \infty$ is strictly increasing, $x_{j+1}/x_j \rightarrow 1$, and $f(x_j) = 0$ for every sufficiently large j . Then f vanishes identically for all sufficiently large x . Consequently any two finite power-logarithm expressions that agree at every sufficiently large integer agree as analytic germs at infinity. They agree on every domain reached by coherent analytic continuation.*

Proof. First combine exponents whose differences are integers. In each such class choose the largest integer translate among the finitely many displayed exponents; this keeps all coefficient functions holomorphic at zero. Unless f is identically zero, in each remaining class there is a first nonzero Taylor coefficient among its finitely many $h_{\nu k}$. Among these finitely many leading terms take the largest real exponent σ , and among terms with that real exponent the largest logarithmic degree d . Convergent Taylor expansion then gives

$$\frac{f(x)}{x^\sigma (\log x)^d} = \sum_{\ell=1}^q c_\ell e^{i\tau_\ell \log x} + o(1), \quad x \rightarrow +\infty, \quad (5.2)$$

where the real τ_ℓ are distinct and at least one c_ℓ is nonzero. The remainder tends to zero on the whole positive ray.

Put $P(t) = \sum c_\ell e^{i\tau_\ell t}$ and $t_j = \log x_j$. The hypothesis gives $P(t_j) \rightarrow 0$ and $t_{j+1} - t_j \rightarrow 0$. The derivative of P is bounded; hence interpolation between adjacent t_j gives $P(t) \rightarrow 0$ as $t \rightarrow \infty$. On the other hand direct integration of the finitely many exponentials shows

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |P(t)|^2 dt = \sum_{\ell=1}^q |c_\ell|^2 > 0.$$

This contradicts $P(t) \rightarrow 0$. Thus the assumed nonzero leading term cannot occur, and the germ is zero. The final assertion follows from the identity theorem. $\square$

The logarithmic-mesh condition includes $x_j = j^q$ for every real $q > 0$. It cannot simply be omitted: the nonzero function

$$\sin\left(\frac{2\pi \log x}{\log 2}\right) = \frac{x^{2\pi i / \log 2} - x^{-2\pi i / \log 2}}{2i}$$

belongs to $\mathcal{P}$ and vanishes at $x = 2^j$. Conversely, if one leaves the finite class entirely, $\sin(\pi x)$ vanishes at every integer. These examples state exactly what is used in the theorem.

Theorem 5.3 (Any number of centres: complete finite reduction). *Let $a_1, \dots, a_d$ be distinct complex numbers and let $\alpha_1, \dots, \alpha_d \in \mathbb{C}$. Use coherent branches of*

$$f(z) = \prod_{j=1}^d (z + a_j)^{\alpha_j}.$$

The following are equivalent:

- (i) *On a nonempty domain, f is a finite constant-coefficient linear combination of single-centre order jets $(z + a_j)^r \text{Log}^k(z + a_j)$, with arbitrary $r \in \mathbb{C}$ and $k \in \mathbb{Z}_{\geq 0}$.*
- (ii) *Such a finite combination agrees with $f(n)$ at every sufficiently large positive integer n .*
- (iii) *Either every α_j is integral, or exactly one α_j is nonintegral and all the other exponents are nonnegative integers.*

The equivalence remains valid if the sampled set in (ii) is any sequence in Theorem 5.2.

Proof. The sampling theorem proves (ii) $\Rightarrow$ (i), and analytic continuation proves the converse after the branches are fixed. If all exponents are integers, ordinary rational partial fractions give (i). If only α_i is nonintegral and all remaining exponents are nonnegative integers, expand their polynomial product in $z + a_i$. This proves (iii) $\Rightarrow$ (i).

For necessity, suppose $\alpha_i \notin \mathbb{Z}$, write $w = z + a_i$, and put

$$h(w) = \prod_{j \neq i} (w + a_j - a_i)^{\alpha_j}.$$

This function is holomorphic near $w = 0$. The monodromy difference $M_i - I$ around a small loop about zero annihilates every single-centre term based at $-a_j$ with $j \neq i$. On the product it gives $(e^{2\pi i \alpha_i} - 1)w^{\alpha_i} h(w)$. The remaining finite sum of $w^r \text{Log}^k w$ is annihilated by a nonzero polynomial $P(E)$ in $E = w d/dw$: include a factor $(E - r)^{k+1}$ for each term. Expanding $h(w) = \sum_{m \geq 0} h_m w^m$ therefore yields

$$P(\alpha_i + m)h_m = 0 \quad (m \geq 0).$$

A polynomial has finitely many roots, so h is a polynomial. Near every distinct point $w = a_i - a_j$, its factor there is a nonzero holomorphic factor times $(w + a_j - a_i)^{\alpha_j}$. Polynomiality forces $\alpha_j \in \mathbb{Z}_{\geq 0}$ for every $j \neq i$. There can be no second nonintegral exponent. This proves necessity. $\square$

Corollary 5.4 (Finite closure of several twisted Fourier families). *For distinct real noninteger twists θ_j , let a periodic distribution K have Fourier coefficients*

$$\widehat{K}(n) = \prod_{j=1}^d (2\pi i(n + \theta_j))^{\alpha_j},$$

where $\text{Log}(2\pi i x) = \log(2\pi |x|) + \pi i \text{sgn}(x)/2$. Then K is a finite linear combination of single-twist families and their spectral order derivatives if and only if condition (iii) of Theorem 5.3 holds. Allowing an additional finite Fourier polynomial does not change the necessity of that condition.

Proof. On the positive Fourier tail, every phase and power of 2π is a constant. Expanding powers of $\log(2\pi(n + \theta_j)) + \pi i/2$ gives precisely the class in Theorem 5.3; a finite Fourier polynomial changes only finitely many modes. This proves necessity. For sufficiency, rational partial fractions and polynomial expansion hold coefficient by coefficient on both tails with the stated branches; only one factor can be nonintegral. The coefficients have locally uniform polynomial growth, so Fourier equality proves distributional equality. $\square$

Thus the two-centre criterion in the source extends to the actual integer Fourier spectrum, and to arbitrarily many centres. The claim concerns a prescribed finite family of functions. It says nothing about rational or algebraic independence of individual constants obtained by evaluating the distributions at particular arguments.

6 Source corrections, verification, and exact claim status

6.1 Corrections and qualifications for integration

The independent-order interpolant needs its established attribution. Any integration of Section 2 should cite Ihara, Nakamura, and Yamamoto for the one-endpoint interpolant, its entire spectral continuation, and its harmonic product law [7]. Equation (2.7) identifies the functions

exactly. This is a literature reconciliation that supplies part of an open repository question, not a newly discovered one-endpoint theorem. The smallest Gamma-separating kernel requested in the full question remains open. The Gamma formulas in Section 2 evaluate symmetric word combinations; they do not evaluate each ordering separately.

Scalar-only qualifications can now be strengthened. The mixed-spectral source carefully restricts its finite two-centre criterion to a scalar functional identity and asks about sampled Fourier coefficients [4]. Theorem 5.2 supplies exactly the missing implication in its finite power–logarithm class. Thus that qualification may now be replaced by Theorem 5.3 and Corollary 5.4. This corrects an incomplete research status, rather than a false existing theorem. No conclusion about individual period independence should be appended.

Local polylogarithm estimates in a specifically inspected preprint. In arXiv:2510.10093v1, equation (2.3) prints a maximum where the local growth estimate requires a minimum, and equation (2.7) includes $s = 1$ in an $O(1)$ case [13]. The elementary counterexamples are

$$\mathrm{Li}_0(e^{-y}) = \frac{1}{e^y - 1} \sim y^{-1}, \quad \mathrm{Li}_1(e^{-y}) = -\log(1 - e^{-y}) = -\log y + O(y).$$

For fixed $r \notin \mathbb{Z}_{>0}$ the corrected first estimate is

$$\mathrm{Li}_r(e^{-y}) = O(y^{\min(0, \Re r - 1)}), \quad y \downarrow 0.$$

For the second, use $O(1 + |\log y|)$ at $s = 1$; the bounded case holds for $\Re s \geq 1$, $s \neq 1$. These statements follow directly from the convergent local formulas (4.16) and Jonquière’s expansion. The subsequent “any $\epsilon > 0$ ” in equation (2.8) should be a sufficiently small positive exponent. For example, its hypotheses permit any

$$0 < \epsilon < \Re t + \min(0, \Re s - 1),$$

which also absorbs the logarithm when $s = 1$. These changes repair the local convergence justification. They do not by themselves refute the paper’s continuation identities. The finding is about this inspected preprint version; the journal version has not been asserted to contain the same printing errors.

Already reported corrections should not be relabeled as new. The directional-zeta package already records the author-PDF corrections $H_0 = 0$, the positive-integer logarithm coefficient in the Borwein–Dilcher local expansion, and the local-series restriction on the splitting point [6]. The present article uses the corrected integer Jonquière formula. It carries those corrections forward by reference instead of claiming a second discovery.

Spectral restriction, finite parts, and color confluence. The Tornheim origin is not a joint regular point. Keep the polar quotients in (3.1) until a direction has been specified. The cubic theorem excludes $a + c = 0$ and $b + c = 0$; it must not be evaluated on either polar hyperplane by cancelling a numerical zero. The mixed-order theorem keeps the outer order C free, so its C -derivatives are valid. It keeps the inner integers fixed, so their transverse derivatives are not supplied. At a resonant color take the meromorphic germ before its Laurent coefficients. Apparent denominators in a nonconfluent partial-fraction expression are handled by the repeated-pole Taylor formula or by the Bernoulli branch of the elimination rule, not by substituting $q = 1$ in (4.13).

6.2 The Gaussian conjectures and the bounded unsuccessful search

At the pinned revision, the relevant canonical labels are `cycloquot:conj:S6` and `s8new:conj:S8` [1]. The earlier rejected S_8 candidate is a different statement; it must not be confused with the revised candidate. The already proved S_4 identity retains its proved status. Nothing in this article proves or refutes the current S_6 or S_8 candidates.

A bounded exploratory S_6 calculation was attempted with the frozen primitive weight-seven target. It generated 51,244 distinct imaginary double-shuffle rows for factor-weight splits (1, 6), (2, 5), and (3, 4), including the common single-divergence cancellation, and two convergent Cayley rows. The system uses 24,514 imaginary word coordinates. Elimination modulo 2, 147, 483, 647 was interrupted before the relation system was exhausted: the last checkpoint had 8,500 pivots, 29,853,917 stored entries, and 393 remaining target terms.

Those numbers report an unfinished computation, not a rank theorem. No row-membership certificate, no nonmembership result, and no separating functional for that enlarged system was obtained. The generator, letter conventions, target, and termination record are included under `exploratory/s6`. Large matrices and compiled executables are omitted because the generator reproduces them. A future successful modular search would still require rational reconstruction and independent exact replay before proving the conjecture. A future negative conclusion would have to specify an exhausted relation space and certify its separating functional.

6.3 Reproducible verification and what it establishes

The archive contains a single runner, `code/verify_all.py`, and separate scripts for the three computational tracks. Exact checks use integers, rational numbers, or exact algebraic expressions. The mixed-order checker evaluates original weighted coefficients by direct convolution and compares them with finite nested sums from the output; it does not reuse the shuffle or elimination code to construct its reference values.

Track	Independent checks	Recorded outcome
Hurwitz interval functions	Finite strict sums, ordered quotient, reversed-star form, residue samples, Bernoulli reduction, and a corrupted-sign control	743 exact assertions; 340 words included
Hurwitz analytic diagnostics	Euler–Maclaurin continuation, singular-intersection Cauchy extraction, shifts, composition, Gamma products, and quadrature	24 diagnostics at 65 digits; largest residual about $3.03 \cdot 10^{-31}$
Cubic Tornheim directions	Symbolic quadratic constraints and cancellation identities; independent subtracted Mellin evaluations on three rays	Exact checks pass; largest numerical discrepancy about $2.27 \cdot 10^{-38}$ at 65 digits
Mixed-order compiler	Original coefficient convolution against output nested sums, including repeated colors, depth four, and weighted roots	192 exact coefficients in 13 varied cases
Explicit triple specializations	Independent elementary kernel expansions, Laurent coefficients, Bernoulli leading poles, Euler primitive, and convergent Mellin integral	24 exact assertions; $C = 4$ residual about $3.89 \cdot 10^{-61}$ at 60 digits

Table 2: Verification scope. Decimal residuals are diagnostics, not interval-certified accuracy claims.

The mathematical proofs establish arbitrary depth, arbitrary allowed directions, and meromorphic

continuation. No finite collection of coefficient checks establishes those generalities by itself. The compiler is a transparent implementation of the finite proof; its algebraic simplifier is not claimed to have optimal complexity. Inputs with floating-point colors or uncertified color domains are rejected rather than silently treated as exact numbers.

The package also records the source commit, hashes of the incoming archives and consulted source files, the individual result ledgers, and the PDF build and visual inspection. The modular TeX sections have distinct label prefixes `eht:`, `tr3:`, `mixed:`, and `sampling:`. Integration notes describe dependencies and conjecture statuses. The standalone TeX is generated from those modular sources so that the PDF can be rebuilt without the ProveIt repository or a network connection.

7 Further exact research questions

The questions below concern identities, finite reductions, and the structure of the remaining analytic coordinates. They are proposed problems, not conjectures asserted on numerical evidence. Each specifies an initial case or a concrete output that could settle the problem.

Question 1 (Evaluate or relate the two cubic coordinates). Find a second functional identity that reduces Ω or κ to a prescribed algebra of ordinary zeta derivatives, Gamma or Barnes values, and elementary constants. The convergent moment in (3.10) and the polylogarithmic integral (3.12) provide two explicit starting points. An equality between those representations is not itself an ordinary-constant evaluation. A useful first target is a reflection or multiplication identity for the whole Dirichlet series $D(A)$, whose second derivative controls κ .

Question 2 (The quartic and higher directional layers). The cubic theorem uses the degree-two Taylor jet of the holomorphic remainder R . At degree four, compute the degree-three jet subject to symmetry, the exact C axis, and the Euler relation (3.5). Give an explicit finite set of convergent Gamma or multiple-Gamma moments spanning all $W_{a,b,c}^{(4)}(0)$. Determine the exact dimension of the formal jet quotient under the proved functional relations. This is a finite linear-algebra question and should be separated from numerical or arithmetic independence of the resulting constants.

Question 3 (Systematic cancellation of directional coordinates). The coefficient of κ vanishes on a conic, and cyclic averaging eliminates it identically. At each higher derivative order, construct the polynomial map from directions to the residual jet coordinates. Compute its kernel for permutation sums and for small finite families of rational directions. The desired output is an exact criterion for combinations reducible to already known coordinates, with the smallest number of directions needed within that specified formal relation space.

Question 4 (The minimal Gamma-separated independent-order kernel). Complete Question 8 of the exact-identities report: beyond the entire interpolant identified here, is there a smallest explicit generating kernel whose singular part separates through a Gamma factor when all inner orders move independently? Begin at depth three near $(s_1, s_2, s_3) = (1, 1, 1)$ and retain three independent perturbations. Use the coefficients $C_{k,m}$ in (2.11) to compare candidate factorizations at every intersecting prefix divisor. State the equivalence relation under which “smallest” is meant.

Question 5 (Nonsymmetric harmonic jets and the coordinate η). Equations (2.24)–(2.25) show precisely where $\eta(a)$ survives after entire normalization. Determine which linear combinations of second and third independent order jets cancel all new ordered coordinates. At rational endpoints, compute a finite character-projection matrix including conductor logarithms. A first target is the antisymmetric second jet at depth two with endpoints $a = 1/2$, $b = 1$; an ordinary Stieltjes reduction would require an additional identity beyond the symmetric Gamma generator.

Question 6 (Reflection and multiplication of normalized primitives). For the primitive (2.29) and its nonpositive-inner extensions, derive exact formulas under $a \mapsto 1 - a$ and $a \mapsto qa$. Keep the base endpoint fixed or explicitly transform it, so additive constants remain determined. Differentiate the whole spectral identity to obtain log-weighted Gamma and Stieltjes primitives. Compare the resulting constants with standard Barnes multiple-Gamma normalizations at the first two derivative orders.

Question 7 (Minimal depth within the mixed-order output class). The bound $P + 1$ is constructive, not optimal in every example. Characterize when a mixed input reduces to depth at most P , or to ordinary Hurwitz functions and their order derivatives. Start with one negative factor and two positive factors at a fixed root-of-unity alphabet. Any minimality theorem must specify the coefficient field, the allowed argument transformations, and the set of functional relations; otherwise a depth count alone is not an obstruction.

Question 8 (A finite map for character and color cancellations). Apply root-of-unity projections to the exact compiler before evaluating periods. Can the positive-depth terms be represented in a finite quotient whose kernel detects all reductions to the depth-one Stieltjes formulas (4.41) at a fixed weight and derivative order? Repeated colors should be treated within the confluent formula, so the reduction matrix remains meaningful when several input colors coincide. An exact row certificate is the appropriate output.

Question 9 (Spectral derivatives beyond locally determined constants). Theorem 4.4 makes every Laurent constant at a nonpositive outer integer a local calculation. The next regular coefficient usually depends on the global Mellin remainder. Classify linear combinations of mixed kernels for which that remainder cancels to one or two prescribed orders. The two explicit triple kernels in Section 4 provide a small initial family; their finite MPL normal forms permit legitimate differentiation before specialization.

Question 10 (Simultaneous color and order resonance). Let several X_j tend to 1 while C tends to a pole and its derivatives are taken. Determine a joint completed germ whose local color logarithms and spectral Laurent coefficients are explicit. The radial variable must respect the weights, $X_j \mapsto \rho^{p_j} X_j$. A first target is one negative and two positive inner orders with two colors approaching 1 at different rates. Distinguish a joint analytic continuation from an ordinary Abel limit of a divergent series.

Question 11 (Four-factor kernels and higher coincident products). Use the mixed-order compiler for the integer-index pieces of the four-factor Fourier zero-frequency constraint. Identify a finite collection of constrained zeta kernels for the different sign partitions, then compute the fully coincident fourth digamma finite part in one fixed endpoint coordinate. The cubic directional formulas should supply boundary checks. The unresolved global kernel must not be replaced by an undefined product of singular distributions.

Question 12 (Sampling sets beyond vanishing logarithmic gaps). Theorem 5.2 is sharp against unrestricted geometric sampling, but a particular set of imaginary exponents may admit much sparser uniqueness sets. For a prescribed finite frequency set $\{\Im \alpha_\nu\}$, characterize the sampling sequences on which its trigonometric leading part is determined. Separate this finite frequency problem from extensions to asymptotic expansions with exponentially small terms, where functions invisible to every power-logarithm coefficient can occur.

Question 13 (Exact Gaussian certificates). Resume S_6 from the frozen rational target and the precisely specified relation space, using a strategy that controls sparse fill and records row provenance. A modular candidate must be lifted and replayed over the rationals. If the chosen space does not contain the target, compute a separating functional for that exhausted space and then identify a

mathematically justified new relation family. Treat the revised S_8 candidate separately; numerical agreement alone cannot promote either to a theorem.

Question 14 (Formal proof kernels and reusable interfaces). Formalize the finite root filter, sparse shuffle, repeated-pole Taylor extraction, geometric/Bernoulli elimination, and polynomial jet maps. Their analytic interface consists of a reusable Mellin-subtraction lemma with locally normal parameter dependence and an explicit reciprocal-Gamma zero. A second interface should formalize prefix residue cancellation at intersecting divisors. These finite and analytic components can then support later repository identities without conflating exact algebra, analytic continuation, and numerical evaluation.

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